



Project

FULL STACK WEB DEVELOPMENT

BSSE- 6B

Presented by:

Tayyab Janjua (231736)

Presented to:

Mr.Shareef Hussain



Github Link for All the tasks: <https://github.com/RajaaTayyab/Full-Stack-Programming-Lab>

Customer Relationship Management (CRM) System

Module 1: Authentication System

Design and implement a secure authentication system using JWT for user access control in the CRM system. The system must ensure that only authenticated users can access protected routes and dashboard features.

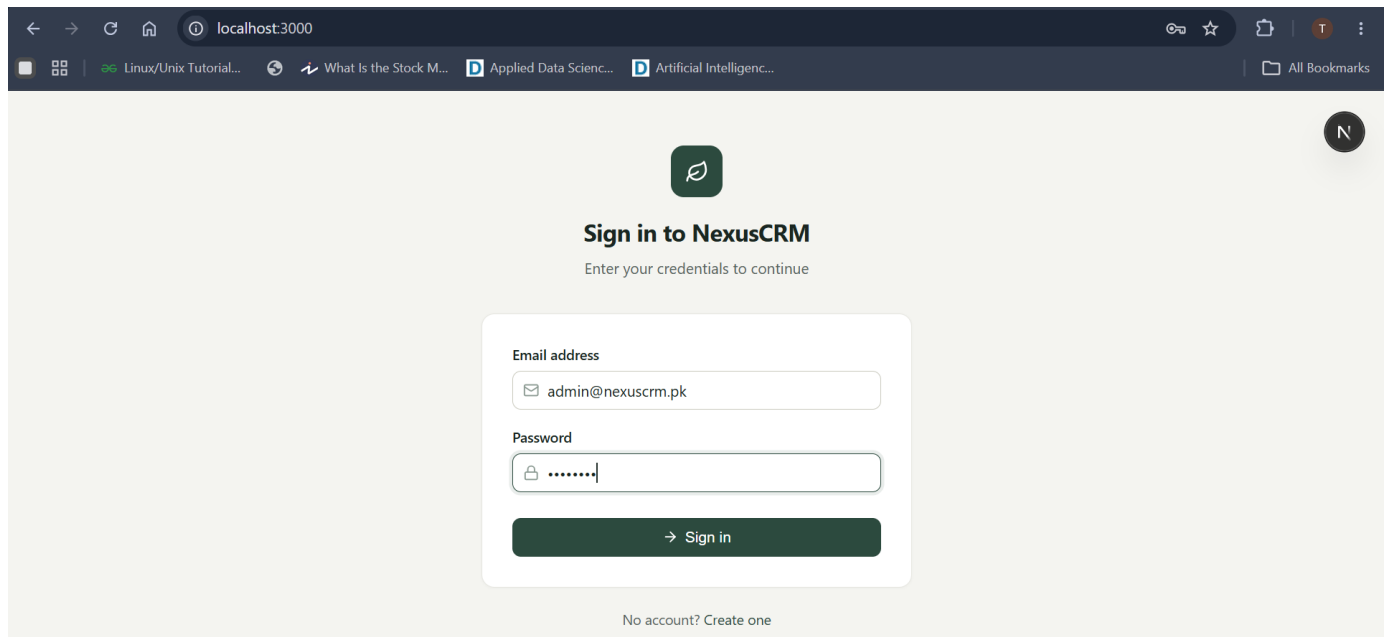
Requirements:

- User registration with name, email, and password
- Password hashing for secure storage
- User login with JWT token generation
- Protected routes for dashboard access
- Logout functionality by clearing session/token
- Restrict unauthorized API access

Account Creation

The screenshot displays a web browser window with the address bar showing 'localhost3000/register'. The page features a dark green logo at the top center, followed by the heading 'Create your account' and the subtext 'Get started with NexusCRM today'. Below this is a white registration form with three input fields: 'Full name' (containing 'Tayyab Ahmed Janjua'), 'Email address' (containing 'tayyab@123.com'), and 'Password' (masked with dots). A dark green button with a white arrow and the text 'Create account' is positioned at the bottom of the form. In the bottom left corner, there is a dark green circle with a white 'N' logo. At the bottom center, a link reads 'Already have an account? Sign in'.

User Login



The screenshot shows a web browser window with the address bar at localhost:3000. The page has a dark header with navigation links: Linux/Unix Tutorial..., What Is the Stock M..., Applied Data Scienc..., and Artificial Intelligenc... There is also an 'All Bookmarks' link. The main content area is light gray and features a dark green circular logo with a white leaf-like icon. Below the logo, the text 'Sign in to NexusCRM' is displayed, followed by the instruction 'Enter your credentials to continue'. A white login form is centered on the page, containing two input fields: 'Email address' with the value 'admin@nexuscrm.pk' and 'Password' with masked characters '.....'. A dark green button with a white arrow and the text 'Sign in' is positioned below the password field. At the bottom of the form, there is a link that says 'No account? Create one'.

JWT Token Generator

```
// Generate JWT token
const token = jwt.sign({ id: user._id }, process.env.JWT_SECRET, { expiresIn: '7d' });
res.status(201).json({ token, user: { id: user._id, name: user.name, email: user.email } });

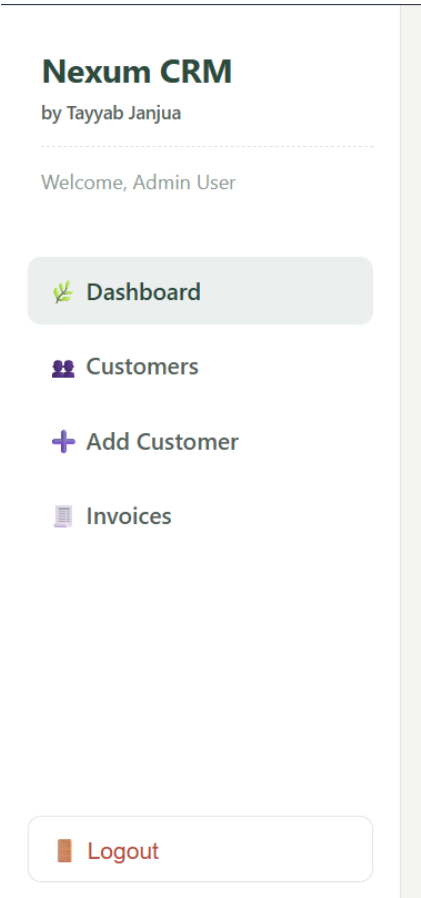
} catch (err) {
  console.error("❌ REGISTRATION ROUTE ERROR:", err);
  res.status(500).json({ message: 'Server error during registration process.', error: err.message });
}
});

module.exports = router;
```

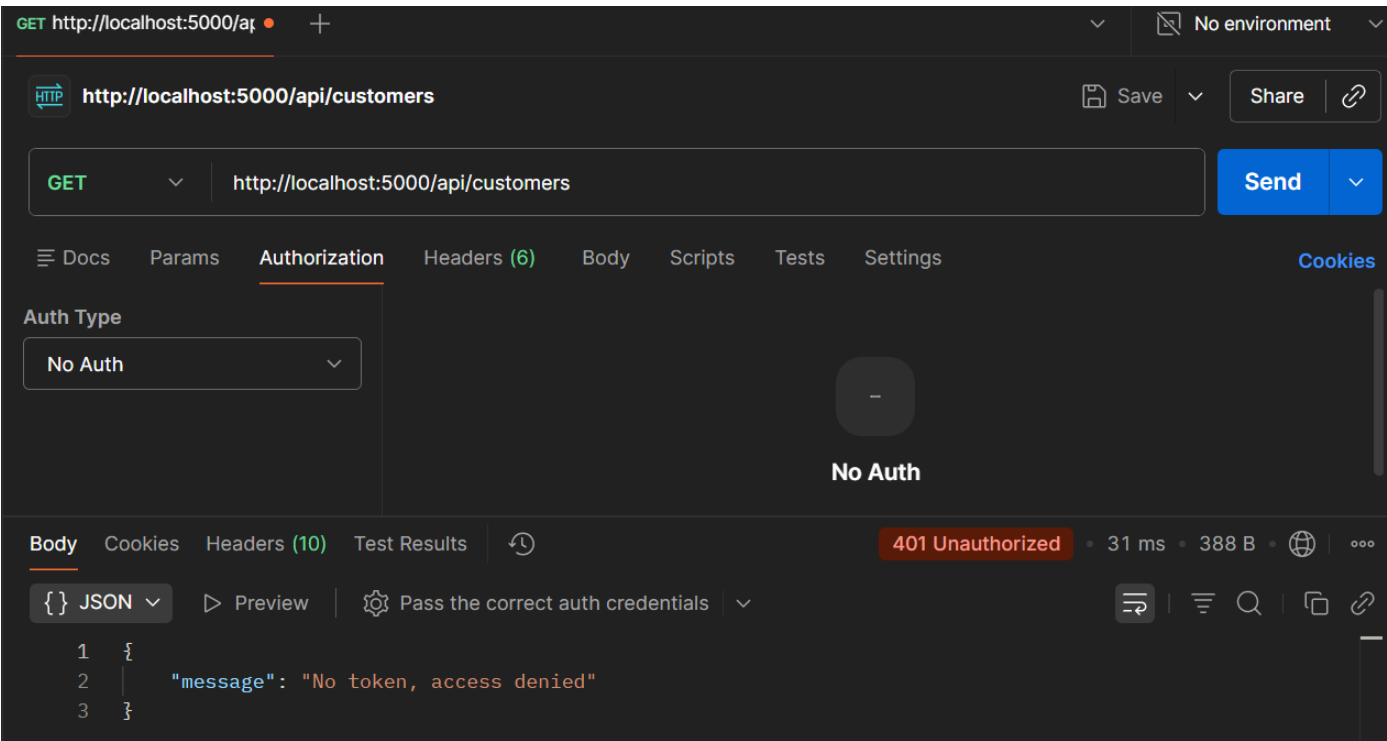
Password Hashing

```
_id: ObjectId('6a21b84f8d17aa6a261edf12')
name: "Admin User"
email: "admin@nexuscrm.pk"
password: "$2b$12$mdnsiNONJAq8mYSb.VdD0eLFzRI8QuVksWaltdcSXC2YzqWfQ84k2"
createdAt: 2026-06-04T17:39:27.448+00:00
```

Logout Functionality



Postman Security Authorization Test



Module 2: Customer Management System

Develop a complete CRUD system for managing customer records. This module represents the core functionality of the CRM system.

Requirements:

- Add new customer records
- View all customers in dashboard
- Update existing customer information
- Delete customer records
- Store data in MongoDB using schema Model
- Insert and manage minimum 15 customer records in the system

Add New Customer

Nexum CRM

by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

+ Add Customer

Invoices

Logout

Add Customer

Fill in the details to add a new customer record

Full Name

Tayyab Ahmed Janjua

Email

tayyab@customer.com

Phone

0331-1234567

Company

Air University

Service

DevOps Engineering

Value (PKR)

25000

Status

Lead

Notes

Additional notes...

Add Customer

Cancel

View All Customers

Nexum CRM

by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

+ Add Customer

Invoices

Logout

Customers

16 total records

Q Search customers...

All

NAME	COMPANY	PHONE	SERVICE	VALUE	STATUS	ACTIONS
Tayyab Ahmed Janjua	Air University	0331-1234567	DevOps Engineering	Rs 25,000	Lead	Edit View Delete
Ahmed Raza	TechCorp PK	0301-1234567	Web Development	Rs 150,000	Active	Edit View Delete
Sara Khan	Digital Hub	0312-2345678	SEO Services	Rs 45,000	Lead	Edit View Delete
Bilal Mahmood	Cloud Nine	0333-3456789	Cloud Hosting	Rs 80,000	Active	Edit View Delete
Ayesha Siddiqui	FashionPK	0321-4567890	E-commerce Setup	Rs 200,000	Inactive	Edit View Delete
Usman Tariq	BuildPro	0345-5678901	CRM Integration	Rs 120,000	Active	Edit View Delete
Zara Mirza	Media Plus	0311-6789012	Social Media Mgmt	Rs 35,000	Lead	Edit View Delete
Hamza Ali	FinTech 360	0322-7890123	Mobile App Dev	Rs 300,000	Active	Edit View Delete

Nexum CRM

by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

+ Add Customer

Invoices

Logout

Customers

16 total records

Q Search customers...

All

Usman Tariq	BuildPro	0345-5678901	CRM Integration	Rs 120,000	Active	Edit View Delete
Zara Mirza	Media Plus	0311-6789012	Social Media Mgmt	Rs 35,000	Lead	Edit View Delete
Hamza Ali	FinTech 360	0322-7890123	Mobile App Dev	Rs 300,000	Active	Edit View Delete
Nadia Sheikh	EduLearn	0334-8901234	LMS Development	Rs 90,000	Lead	Edit View Delete
Faisal Qureshi	RetailMax	0344-9012345	POS System	Rs 175,000	Active	Edit View Delete
Mariam Javed	HealthPlus	0315-0123456	Healthcare Portal	Rs 250,000	Inactive	Edit View Delete
Kashif Nawaz	Logistics PK	0336-1234560	Tracking System	Rs 140,000	Active	Edit View Delete
Sana Riaz	Startup Lab	0347-2345671	MVP Development	Rs 60,000	Lead	Edit View Delete
Tariq Hassan	Exporters PK	0323-3456782	ERP Integration	Rs 380,000	Active	Edit View Delete
Hira Baig	Creative Studio	0352-4567893	UI/UX Design	Rs 55,000	Lead	Edit View Delete
Omar Farooq	SecureTech	0317-5678904	Cybersecurity Audit	Rs 220,000	Active	Edit View Delete

Update Existing Customer

Nexum CRM

by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

Add Customer

Customers

16 total records

Search customers...

All

NAME	COMPANY	PHONE	SERVICE	VALUE	STATUS	ACTIONS
Tayyab Ahmed Janjua	Air University	0331-1234567	DevOps Engineering	Rs 25,000	Lead	✎ 📄 🗑
Ahmed Raza	TechCorp PK	0301-1234567	Web Development	Rs 150,000	Active	✎ 📄 🗑

Nexum CRM

by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

Add Customer

Invoices

Logout

Edit Customer

Full Name

Ahmed Raza

Email

ahmed.raza@techcorp.pk

Phone

0301-1234567

Company

TechCorp PK

Service

Web Development

Value (PKR)

150000

Status

Active

Notes

Long-term client

Save Changes

Cancel

Nexum CRM

by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

Add Customer

Invoices

Logout

Edit Customer

Full Name

Ali Raza

Email

ali.raza@techcorp.pk

Phone

0301-1234567

Company

Jazz

Service

Web Development

Value (PKR)

150000

Status

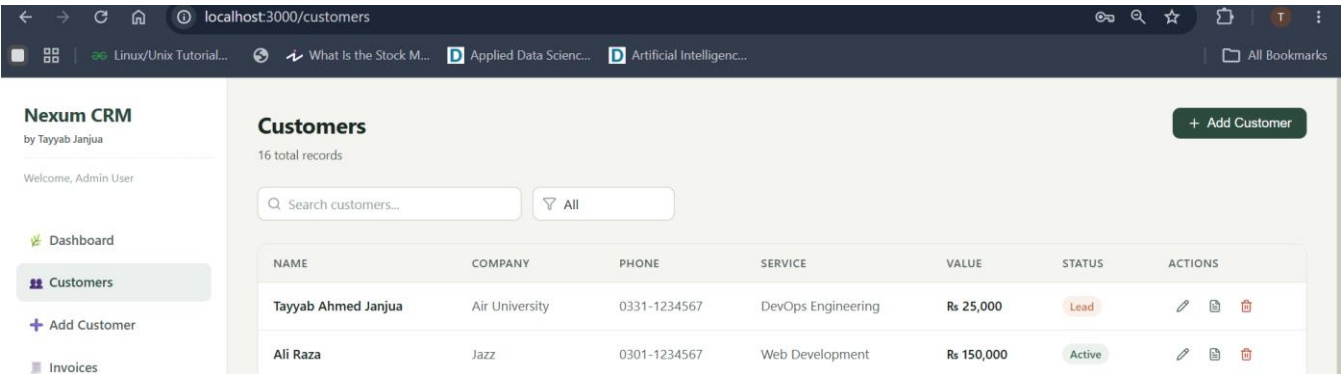
Active

Notes

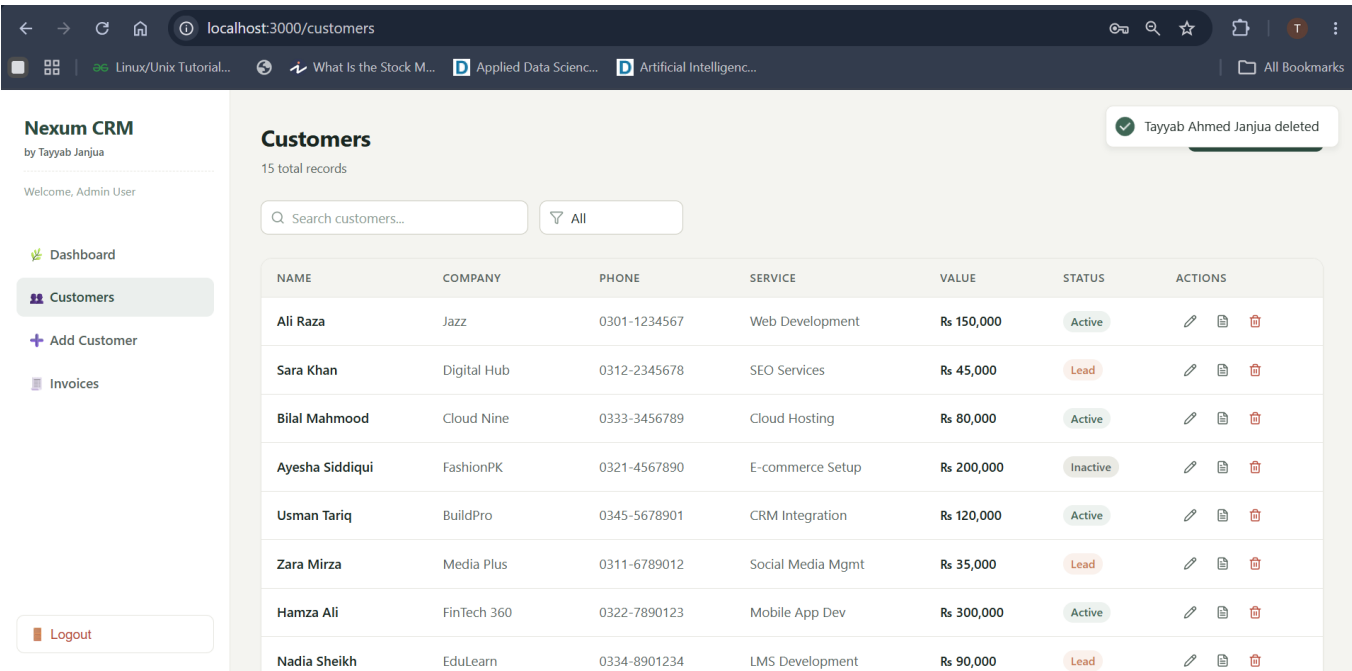
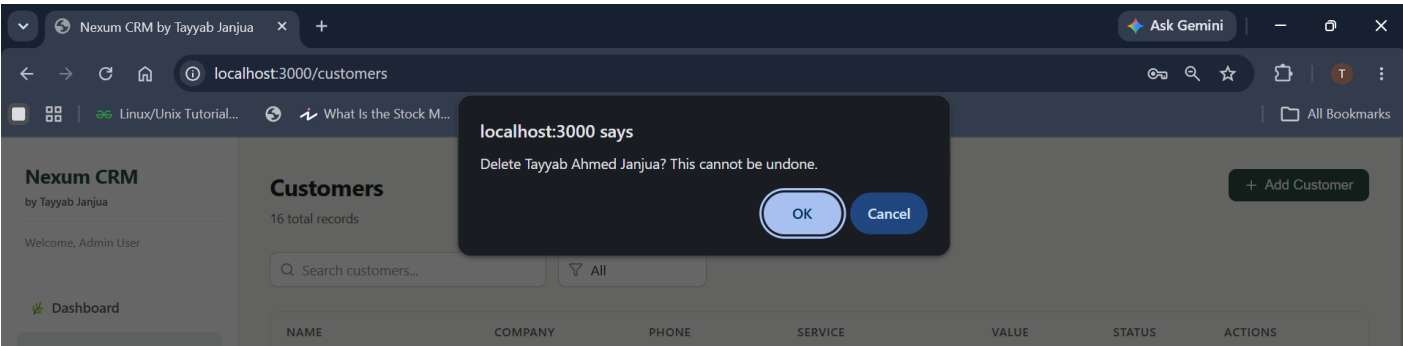
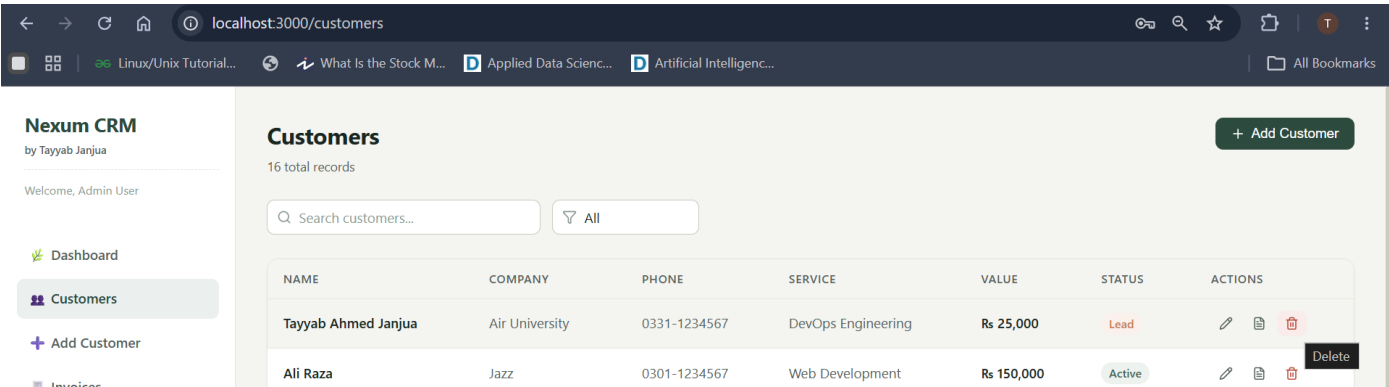
Long-term client

Save Changes

Cancel



Delete Customer



Mongo Db Schema Model

```
Final_Term_Project_CRM > backend > models > JS Customer.js > ...  
1  const mongoose = require('mongoose');  
2  
3  const customerSchema = new mongoose.Schema({  
4    name: { type: String, required: true, trim: true },  
5    email: { type: String, required: true, unique: true, lowercase: true },  
6    phone: { type: String, required: true },  
7    company: { type: String, required: true },  
8    status: { type: String, enum: ['Lead', 'Active', 'Inactive'], default: 'Lead' },  
9    service: { type: String, required: true },  
10   value: { type: Number, required: true, min: 0 },  
11   notes: { type: String, default: '' },  
12   createdAt: { type: Date, default: Date.now }  
13 });  
14  
15 module.exports = mongoose.model('Customer', customerSchema);
```

Mongo Db Data Stored 15 Entries

MongoDB Compass - crm/nexuscrm.customers

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

crm

- admin
- config
- local
- nexuscrm
 - customers
 - invoices
 - users

crm > nexuscrm > customers

Documents 16 Aggregations Schema Indexes 2 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

25 1 - 16 of 16

Document 1

```
{  
  "_id": ObjectId("6a21b84f8d17aa6a261edf13"),  
  "name": "Ahmed Raza",  
  "email": "ahmed.raza@techcorp.pk",  
  "phone": "0301-1234567",  
  "company": "TechCorp PK",  
  "status": "Active",  
  "service": "Web Development",  
  "value": 150000,  
  "notes": "Long-term client",  
  "createdAt": 2026-06-04T17:39:27.514+00:00  
}
```

Document 2

```
{  
  "_id": ObjectId("6a21b84f8d17aa6a261edf14"),  
  "name": "Sara Khan",  
  "email": "sara.khan@digitalhub.io",  
  "phone": "0312-2345678",  
  "company": "Digital Hub",  
  "status": "Lead",  
  "service": "SEO Services",  
  "value": 45000  
}
```

Module 3: Search and Filter System

Implement search and filtering functionality to improve data handling within the CRM system. This feature helps users quickly locate and organize customer records.

Requirements:

- Search customers by name
- Filter customers by status (Lead / Active / Inactive)
- Display dynamic results without page reloading
- Apply functionality on all 15 records

Search Customer by Name

Nexum CRM

by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

+ Add Customer

Invoices

Logout

Customers

1 total records

Om

All

NAME	COMPANY	PHONE	SERVICE	VALUE	STATUS	ACTIONS
Omar Farooq	SecureTech	0317-5678904	Cybersecurity Audit	Rs 220,000	Active	

Filter Customer by Status

Nexum CRM

by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

+ Add Customer

Invoices

Customers

5 total records

Search customers...

Lead

NAME	COMPANY	PHONE	SERVICE	VALUE	STATUS	ACTIONS
Sara Khan	Digital Hub	0312-2345678	SEO Services	Rs 45,000	Lead	
Zara Mirza	Media Plus	0311-6789012	Social Media Mgmt	Rs 35,000	Lead	
Nadia Sheikh	EduLearn	0334-8901234	LMS Development	Rs 90,000	Lead	
Sana Riaz	Startup Lab	0347-2345671	MVP Development	Rs 60,000	Lead	
Hira Baig	Creative Studio	0352-4567893	UI/UX Design	Rs 55,000	Lead	

Functionality Applied on All Records

Nexum CRM

by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

+ Add Customer

Invoices

Logout

Customers

8 total records

Search customers...

Active

NAME	COMPANY	PHONE	SERVICE	VALUE	STATUS	ACTIONS
Ali Raza	Jazz	0301-1234567	Web Development	Rs 150,000	Active	Edit View Delete
Bilal Mahmood	Cloud Nine	0333-3456789	Cloud Hosting	Rs 80,000	Active	Edit View Delete
Usman Tariq	BuildPro	0345-5678901	CRM Integration	Rs 120,000	Active	Edit View Delete
Hamza Ali	FinTech 360	0322-7890123	Mobile App Dev	Rs 300,000	Active	Edit View Delete
Faisal Qureshi	RetailMax	0344-9012345	POS System	Rs 175,000	Active	Edit View Delete
Kashif Nawaz	Logistics PK	0336-1234560	Tracking System	Rs 140,000	Active	Edit View Delete
Tariq Hassan	Exporters PK	0323-3456782	ERP Integration	Rs 380,000	Active	Edit View Delete

Dynamically displaying Results without Refreshing

Nexum CRM

by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

+ Add Customer

Invoices

Logout

Customers

2 total records

Search customers...

Inactive

NAME	COMPANY	PHONE	SERVICE	VALUE	STATUS	ACTIONS
Ayesha Siddiqui	FashionPK	0321-4567890	E-commerce Setup	Rs 200,000	Inactive	Edit View Delete
Mariam Javed	HealthPlus	0315-0123456	Healthcare Portal	Rs 250,000	Inactive	Edit View Delete

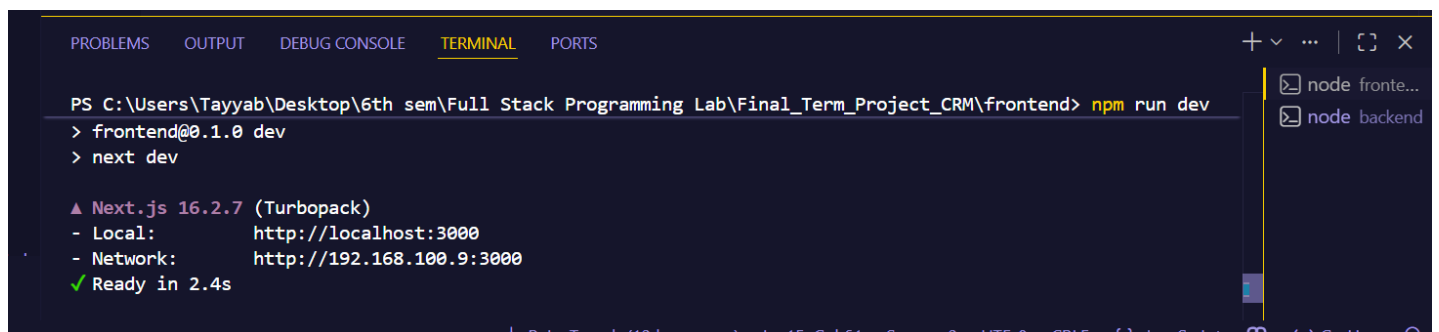
Module 4: Next.js Frontend Application

Develop a modern and responsive frontend using Next.js for the CRM system. The frontend should provide an interactive and user-friendly interface for all operations.

Requirements:

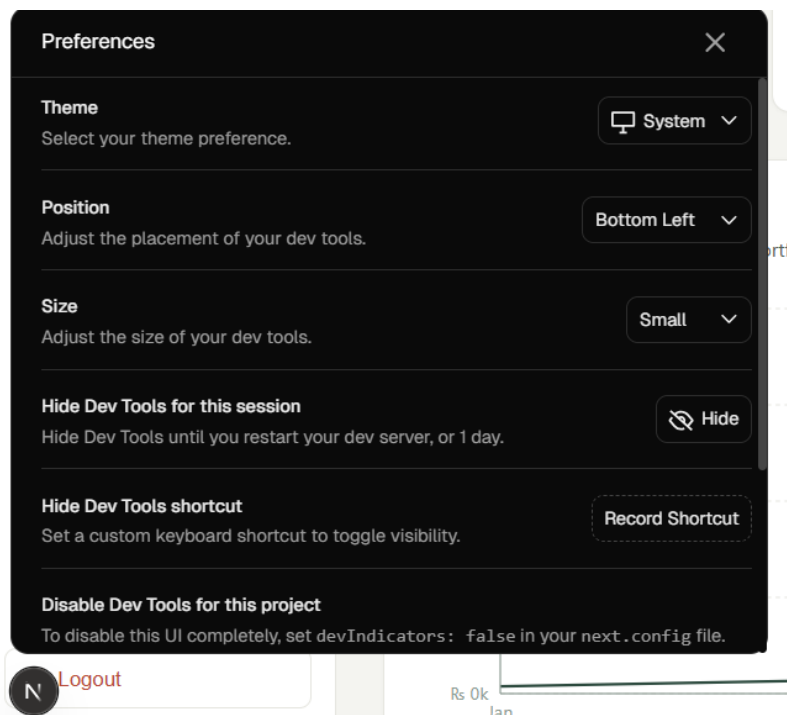
- Login and registration pages
- CRM dashboard interface
- Add and edit customer forms
- Protected routing system
- Reusable React components
- API integration using Axios or Fetch

Next Js System



```
PS C:\Users\Tayyab\Desktop\6th sem\Full Stack Programming Lab\Final_Term_Project_CRM\frontend> npm run dev
> frontend@0.1.0 dev
> next dev

▲ Next.js 16.2.7 (Turbopack)
- Local:      http://localhost:3000
- Network:    http://192.168.100.9:3000
✓ Ready in 2.4s
```



Api integration using Axios

```
Final_Term_Project_CRM > frontend > utils > JS api.js > [🔗] default
1  import axios from 'axios';
2
3  const api = axios.create({ baseURL: 'http://localhost:5000/api' });
4
5  api.interceptors.request.use((config) => {
6    const token = localStorage.getItem('nexus_token');
7    if (token) config.headers.Authorization = `Bearer ${token}`;
8    return config;
9  });
10
11 export default api;
```

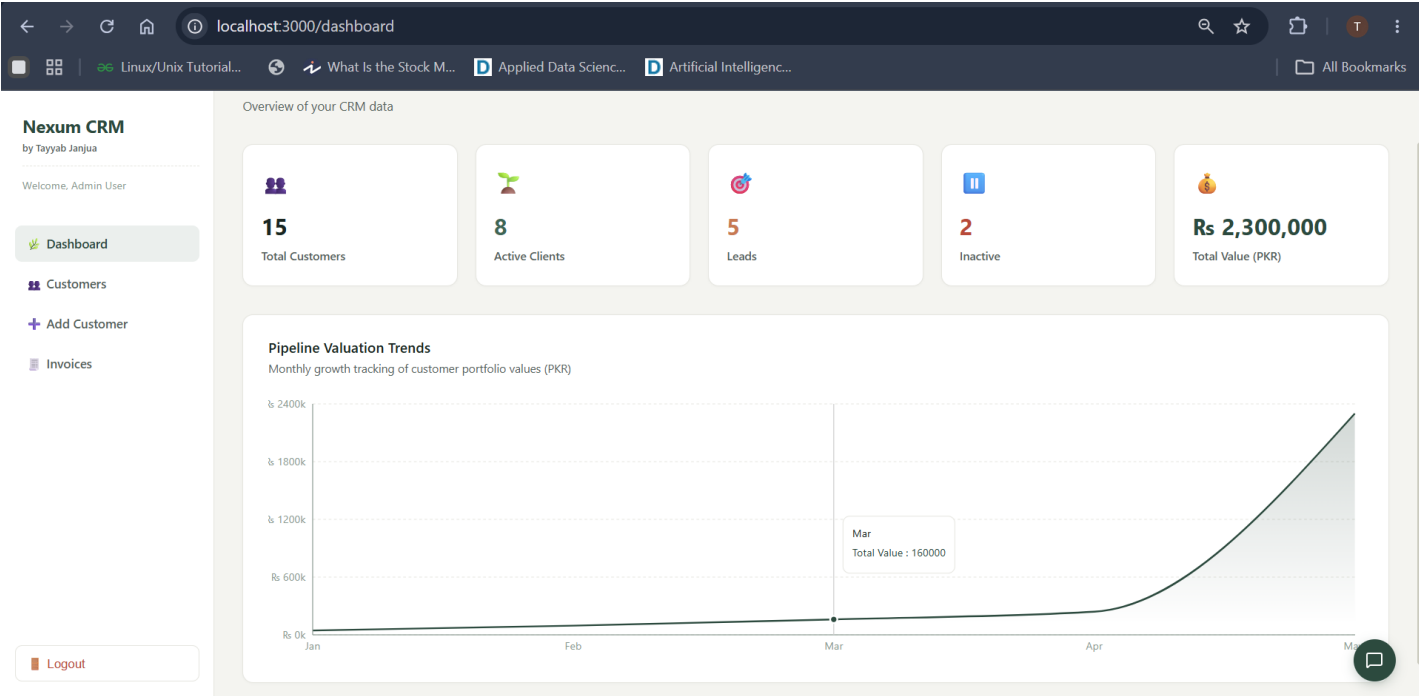
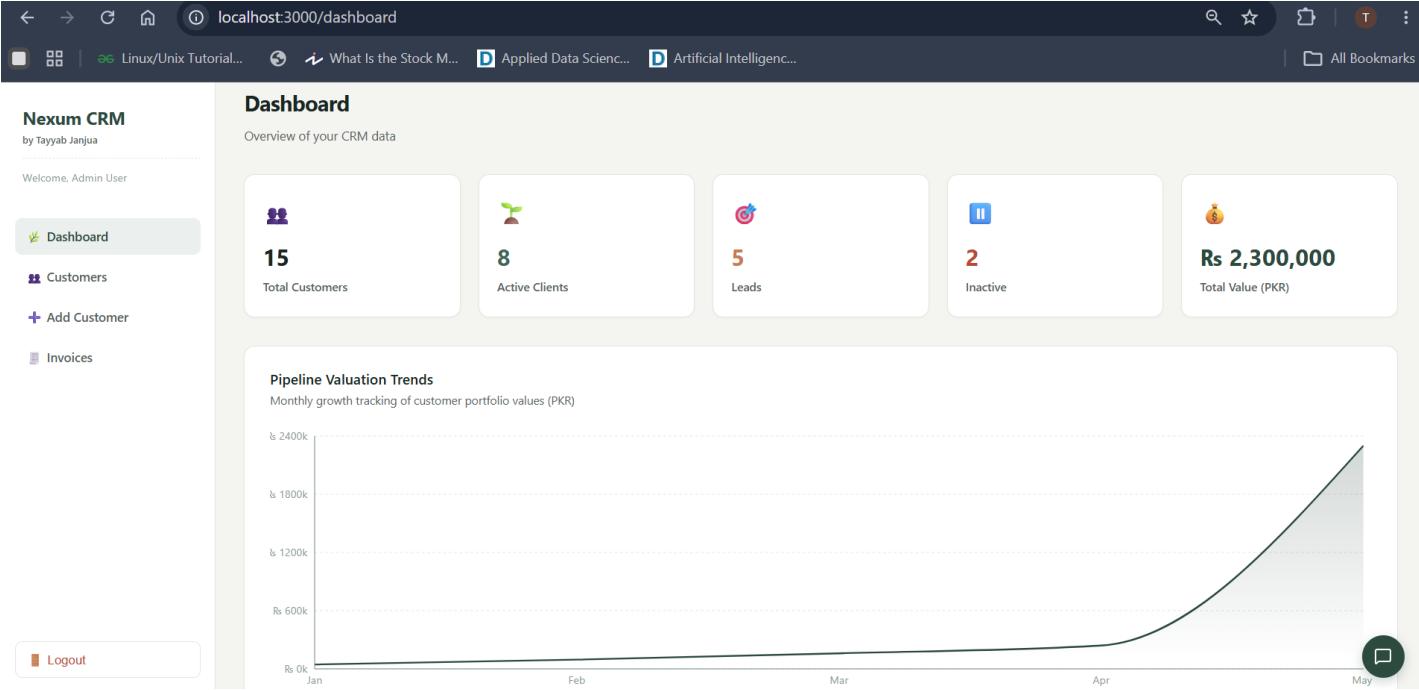
Reusable React Components

```

✓ frontend
  > .next
  ✓ components
    JS Chatbot.js      M
    JS Layout.js       U
```

```
Final_Term_Project_CRM > frontend > components > JS Layout.js > [📁] Layout
1  import { useAuth } from '../context/AuthContext';
2  import { useRouter } from 'next/router';
3  import { useEffect } from 'react';
4  import Link from 'next/link';
5
6  export default function Layout({ children }) {
7    const { user, logout, loading } = useAuth();
8    const router = useRouter();
9
10    useEffect(() => {
11      if (!loading && !user) router.push('/');
12    }, [user, loading]);
13
```

CRM Dashboard Interface



Module 5: Invoice Generation System

Implement a basic invoice generation module for customer transactions. This module simulates real-world business billing functionality.

Requirements:

- Generate invoices for selected customers
- Display invoice details (customer, services, total amount)
- Include date and summary information
- Download invoice as PDF file
- Optional storage in database

Generate Invoice

localhost:3000/invoices/generate/6a21b84f8d17aa6a261edf13

Linux/Unix Tutorial... What Is the Stock M... Applied Data Scienc... Artificial Intelligenc...

Nexum CRM
by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

+ Add Customer

Invoices

Logout

Generate Invoice

For: **Ali Raza** — Jazz

SaaS License Subscription 25000

+ Add Service

Total Amount **Rs 25,000**

Generate Invoice Back

Invoice List Download Button

Nexum CRM

by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

Add Customer

Invoices

Logout

Invoices

Go to Customers → click Invoice to generate one

INVOICE NO	CUSTOMER	SERVICES	TOTAL (PKR)	DATE	ACTION
INV-586153	Sara Khan	Medical	Rs 15,000	6/5/2026	PDF
INV-614091	Ali Raza	SaaS License Subscription	Rs 25,000	6/5/2026	PDF

Invoice_INV-614091.pdf
4.2 KB • Done

PDF for INV-614091 downloaded!

Invoice Summary with Date

File C:/Users/Tayyab/Downloads/Invoice_INV-614091.pdf

Invoice_INV-614091.pdf

1 / 1 | 100%

Nexum CRM

by Tayyab Janjua

Invoice No: INV-614091

Date: 6/5/2026

Customer Reference: Ali Raza

Billed Service Items & Records:

SaaS License Subscription

Rs 25,000

Total Balanced Amount: Rs 25,000

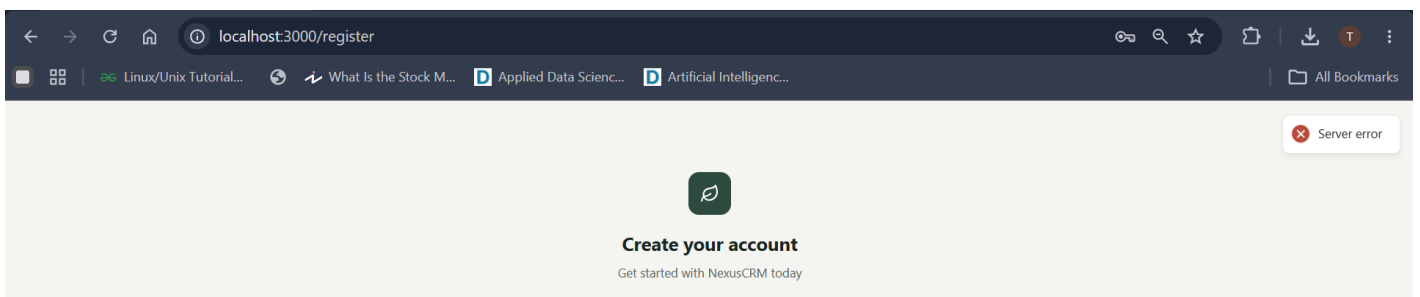
Module 6: Notification System

Implement a notification system to enhance user experience during operations. It provides feedback for all major actions performed in the system.

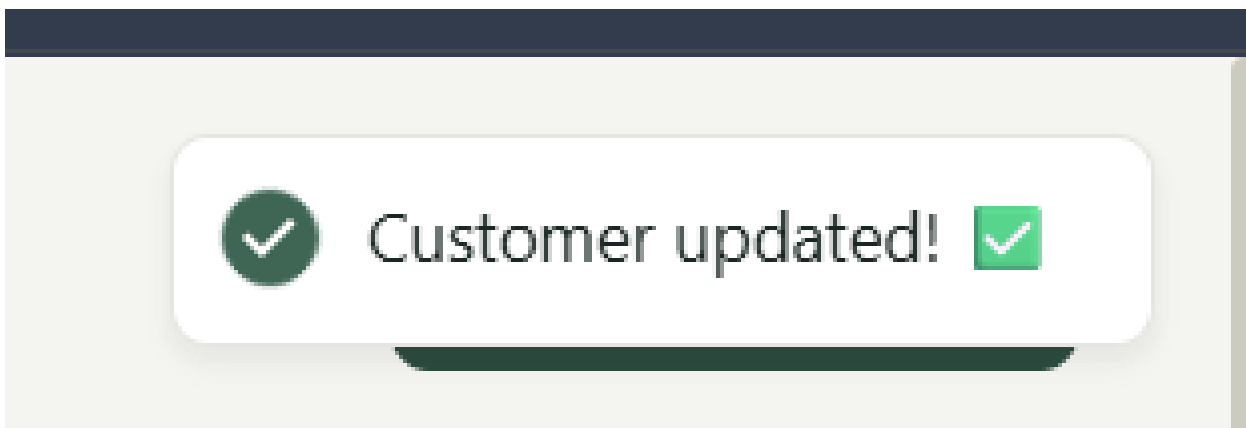
Requirements:

- Show success messages on customer creation
- Show update and delete confirmations
- Display error messages for failed operations
- Use toast or alert-based notifications

Error Message



Success Message



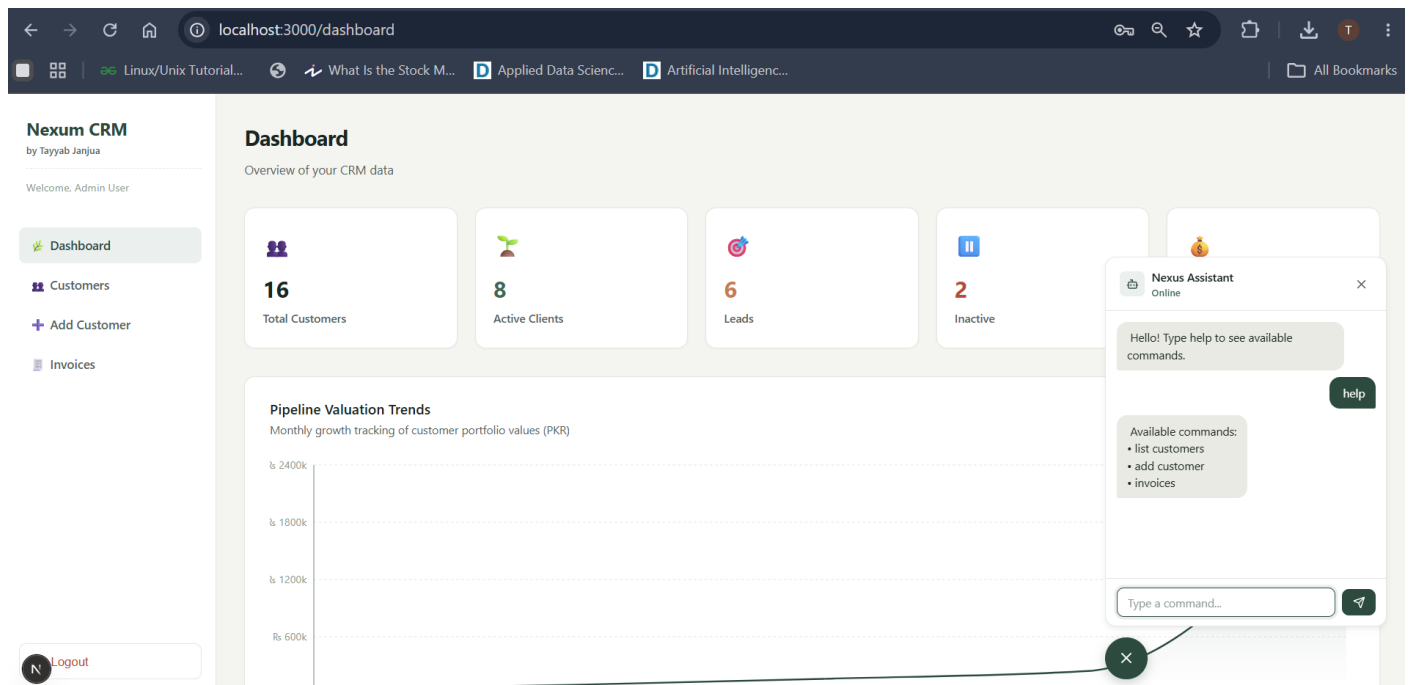
Module 7: Chatbot System

Develop a simple rule-based chatbot inside the CRM dashboard. The chatbot assists users with basic navigation and system commands.

Requirements:

- Respond to predefined commands only
- Show customer list on request
- Navigate to add customer page
- Open invoice generation module
- No external AI API usage allowed

Chatbot Commands



List Customers

Nexum CRM

by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

Add Customer

Invoices

Logout

Dashboard

Overview of your CRM data

16

Total Customers

8

Active Clients

6

Leads

2

Inactive

Rs 2,310,000

Pipeline Valuation Trends

Monthly growth tracking of customer portfolio values (PKR)

Rs 2400k

Rs 1800k

Rs 1200k

Rs 600k

Rs 0k

Jan

Feb

Mar

Apr

May

Nexus Assistant

Online

Available commands:

list customers

add customer

invoices

list customers

Found 16 customers:

hassan (Lead)

Ali Raza (Active)

Sara Khan (Lead)

Bilal Mahmood (Active)

Ayesha Siddiqui (Inactive)

...and 11 more

type a command...

Add Customers

Nexum CRM

by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

Add Customer

Invoices

Logout

Dashboard

Overview of your CRM data

16

Total Customers

8

Active Clients

6

Leads

2

Inactive

Rs 2,310,000

Pipeline Valuation Trends

Monthly growth tracking of customer portfolio values (PKR)

Rs 2400k

Rs 1800k

Rs 1200k

Rs 600k

Rs 0k

Jan

Feb

Mar

Apr

May

Nexus Assistant

Online

Hello! Type help to see available commands.

help

Available commands:

list customers

add customer

invoices

add customer

Nexum CRM

by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

+ Add Customer

Invoices

Logout

Add Customer

Fill in the details to add a new customer record

Full Name

Ahmed Raza

Email

ahmed@example.com

Phone

0301-1234567

Company

Tech Corp

Service

Web Development

Value (PKR)

150000

Status

Lead

Notes

Additional notes...

Add Customer

Cancel

Open Invoice Generation

Nexum CRM

by Tayyab Janjua

Welcome, Admin User

Dashboard

Customers

+ Add Customer

Invoices

Logout

16

Total Customers

8

Active Clients

6

Leads

2

Inactive

Rs 2,310,000

Total Value (PKR)

Pipeline Valuation Trends

Monthly growth tracking of customer portfolio values (PKR)

Rs 2400k

Rs 1800k

Rs 1200k

Rs 600k

Rs 0k

Jan

Feb

Mar

Apr

Nexus Assistant

Online

Hello! Type help to see available commands.

help

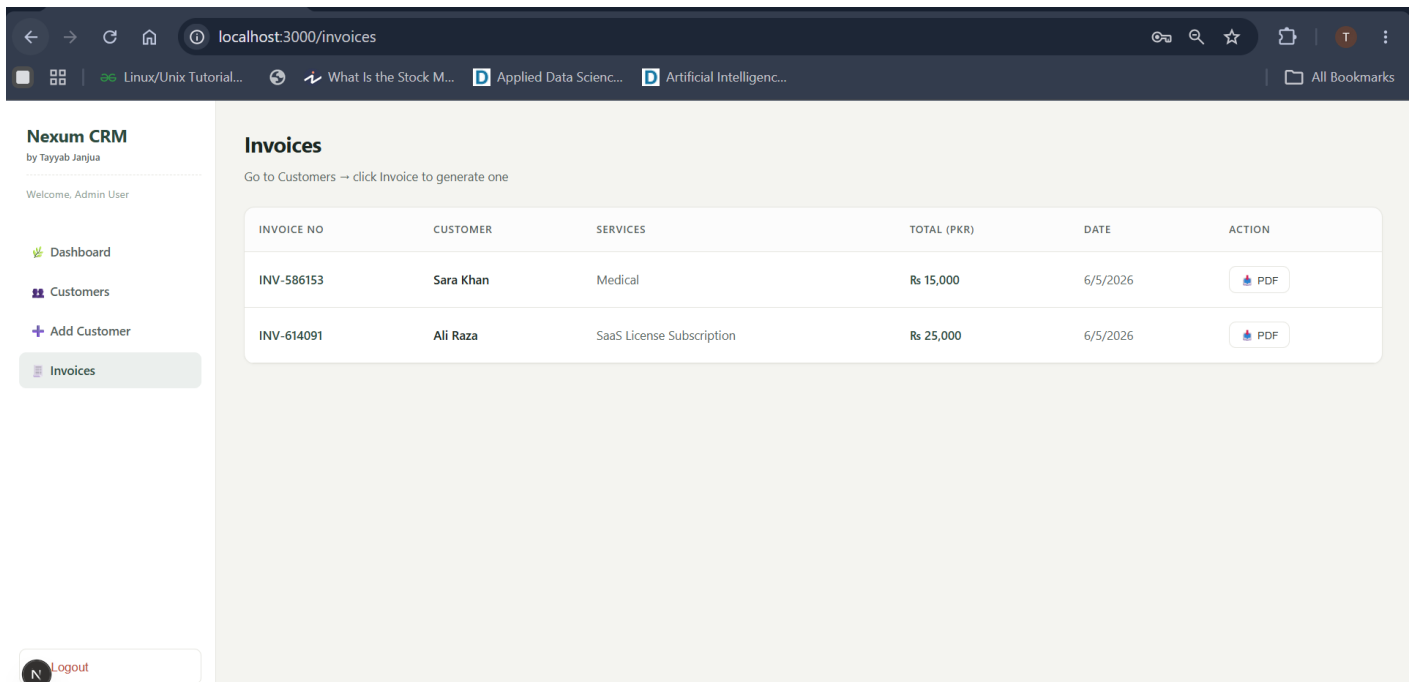
Available commands:

• list customers

• add customer

• invoices

invoices



No external Api Usage

```
Final_Term_Project_CRM > frontend > components > JS Chatbot.js > Chatbot
1 import { useState } from 'react';
2 import { useRouter } from 'next/router';
3 import { MessageSquare, X, Send, Bot } from 'lucide-react';
4 import api from 'utils/api';
5 + function Chatbot(): JSX.Element
6 export default function Chatbot() {
7   const [open, setOpen] = useState(false);
8   const [messages, setMessages] = useState([{ from: 'bot', text: 'Hello! Type help to see available commands.' }]);
9   const [input, setInput] = useState('');
10  const router = useRouter();
```

Module 8: UI/ Code Quality

The system will be evaluated based on user interface and code structure quality. A clean, maintainable, and modular codebase is expected.

Requirements:

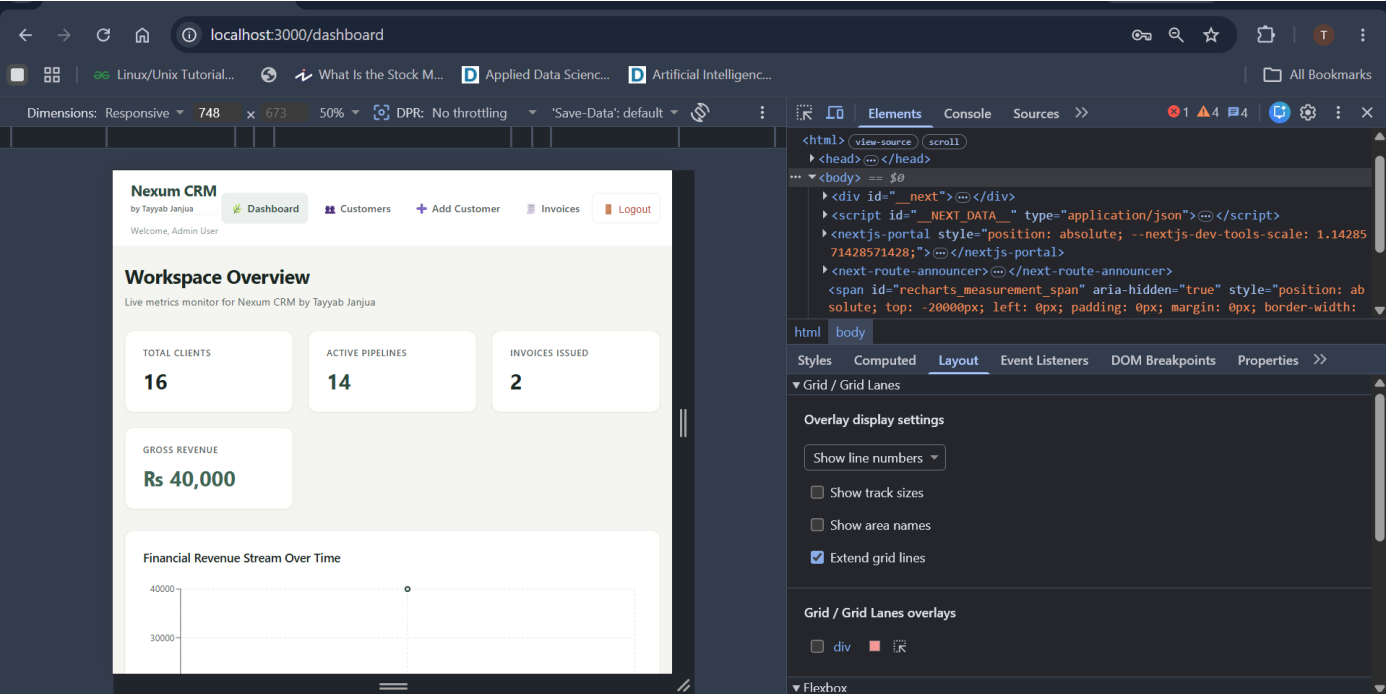
- Clean and responsive UI design
- Proper project folder structure
- Reusable components
- Readable and well-organized code
- Consistent naming conventions

Project Structure with Consistent Naming Conventions

```
Final_Term_Project_CRM/  
├── backend/  
│   ├── config/  
│   │   └── db.js  
│   ├── middleware/  
│   │   └── auth.js  
│   ├── models/  
│   │   ├── Customer.js  
│   │   ├── Invoice.js  
│   │   └── User.js  
│   ├── routes/  
│   │   ├── auth.js  
│   │   ├── customers.js  
│   │   └── invoices.js  
│   ├── .env  
│   ├── package.json  
│   └── server.js  
├── frontend/  
│   ├── components/  
│   │   ├── Chatbot.js  
│   │   ├── Layout.js  
│   │   └── Navbar.js  
│   ├── context/  
│   │   └── AuthContext.js  
│   ├── pages/  
│   │   ├── invoices/  
│   │   │   ├── generate/  
│   │   │   │   └── [id].js  
│   │   │   └── index.js  
│   │   ├── customers.js  
│   │   ├── dashboard.js  
│   │   ├── login.js  
│   │   └── register.js  
│   ├── public/  
│   │   └── favicon.ico  
│   ├── styles/  
│   │   └── globals.css  
│   ├── utils/  
│   │   └── api.js  
│   ├── package.json  
│   └── next.config.js
```

```
Final_Term_Project_C...  
└── Final_Term_Project_C...  
    ├── backend  
    │   ├── middleware  
    │   ├── models  
    │   ├── node_modules  
    │   ├── routes  
    │   ├── .env  
    │   ├── package-lock.json  
    │   ├── package.json  
    │   ├── seed.js  
    │   ├── server.js  
    ├── frontend  
    │   ├── .next  
    │   ├── components  
    │   │   ├── JS Chatbot.js  
    │   │   ├── JS Layout.js  
    │   │   ├── context  
    │   │   ├── node_modules  
    │   │   ├── pages  
    │   │   ├── public  
    │   │   ├── styles  
    │   │   ├── utils  
    │   ├── .gitignore
```

Responsive Page



Clean UI Design

